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TrendOne Customer Analytics



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Project Overview

TrendOne Customer Analytics

An end-to-end customer analytics project exploring shopping behavior, statistical relationships, and customer profiles using a structured customer shopping dataset.

Deliverables

Python Analysis · Statistical Testing · SQL Analysis · Customer Segmentation · Power BI Dashboard

Project Objectives

Analyze customer purchasing patterns and group differences.
Examine relationships between subscription status and discount usage.
Develop exploratory customer segments using K-Means clustering.
Translate analytical findings into business recommendations through a Power BI dashboard.

Project Context

TrendOne is a fictional retail brand used to frame the analysis in a business context. The findings are based on the available dataset and do not represent verified business results from a real company.

Business Problem & Questions



Purchasing Behavior

Do recorded purchase amounts differ across demographic groups, product categories, discount usage, and subscription status?



Customer Engagement

How strongly are subscription status and discount usage associated?



Customer Segementation

Can customers be grouped into exploratory profiles using selected numerical features?

Data & Analytical Workflow

Dataset Overview

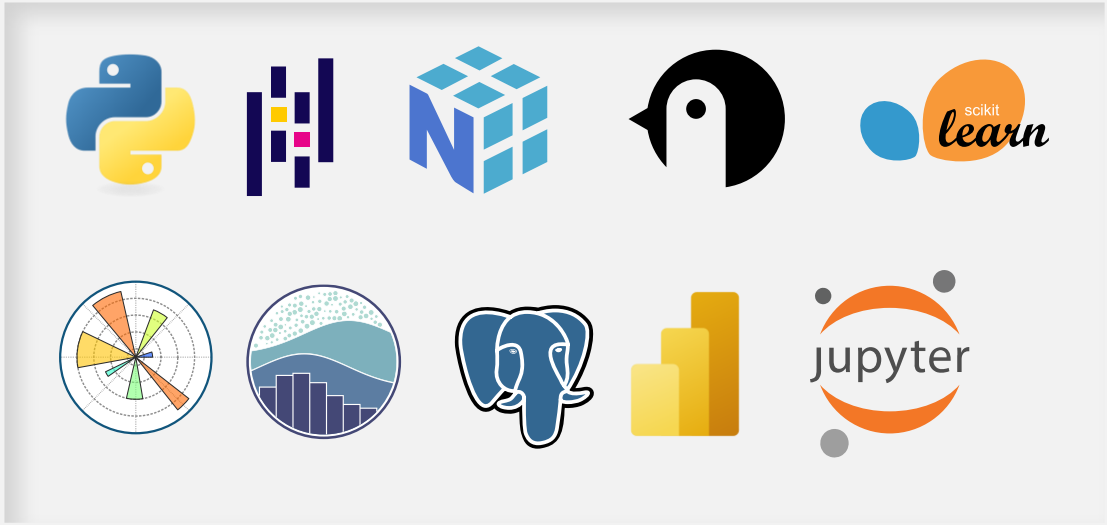
- 3,900 customer records
- 18 original variables
- One record per customer
- 37 missing values in Review Rating (0.95%)
- No exact duplicate rows identified

Data Consideration

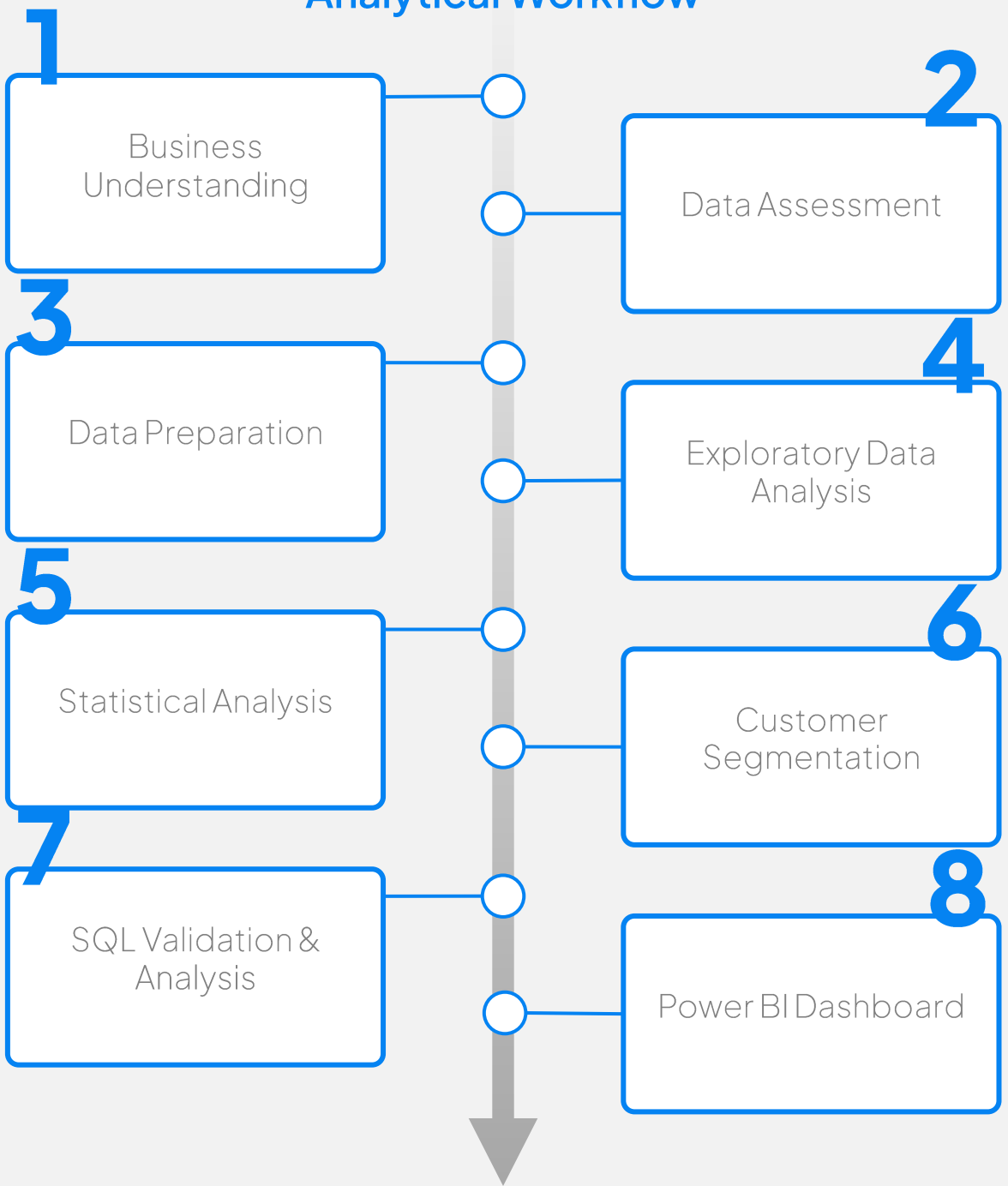
The dataset represents customer-level records rather than transaction-level history. Findings describe patterns in the available dataset and should not be interpreted as proof of causality or actual customer retention.

Tools Used

Python: Pandas, NumPy, Pingouin, SciPy, Scikit-learn, Matplotlib, Seaborn
SQL & Database: PostgreSQL
Data Visualization: Power BI
Development Environment: Jupyter Notebook



Analytical Workflow



Statistical Analysis & Key Findings

No Meaningful Purchase Amount Differences Detected

The analysis found no statistically significant differences in recorded purchase amount across the tested demographic group product categories, discount usage groups, or subscription groups.

Age vs. Purchase Amount

Spearman
$\rho = -0.01$,
$p = 0.510$

Gender vs. Purchase Amount

Welch's t-test,
$p = 0.380$,
Cohen's ,
$d = -0.030$

Product Category vs. Purchase Amount

Welch's ANOVA,
$p = 0.254$,

partial $\eta^2 = 0.0011$

Subscription Status vs. Purchase Amount

Welch's t-test,
$p = 0.660$
Cohen's
$d = -0.016$

Strong Association Between Subscription and Discount Usage

Chi-square test: $\chi^2 = 1{,}908.92$, $p < 0.001$
Cramér's $V = 0.700$

Interpretation

Subscription status and discount usage are strongly associated in this dataset. This does not establish causation, and the underlying data pattern should be investigated further.

Statistical Note

A non-significant result does not prove that groups are identical. Effect sizes were considered alongside p-values to assess the magnitude of observed differences.

Exploratory Customer Segmentation

Objective

Explore customer profiles using K-Means clustering based on selected numerical features.

Model Configuration

- Number of clusters: 6
- Random state: 42
- Initialization runs (n_init): 20
- Silhouette score: 0.2864

Cluster	Exploratory Profile	Avg. Purchase Amount	Avg. Previous Purchases
C0	Higher-Spend, Lower Purchase History	$76.00	12.11
C1	Repeat-History, Lower-Rating	$45.99	38.88
C2	Lower-Spend, Lower Purchase History	$39.48	11.90
C3	Higher-Spend, Higher Purchase History	$80.33	39.76
C4	Higher-Spend, Lower-Rating	$82.77	20.78
C5	Lower-Spend, Higher-Rating	$35.84	28.39

Cluster Profiles

Key Observations

- C4 has the highest average recorded purchase amount but the lowest average rating.
- C3 combines a high average purchase amount with the highest average previous-purchase count.
- C5 has the lowest average recorded purchase amount but the highest average rating.

Interpretation & Caveat

The six clusters are exploratory profiles, not validated customer segments. The silhouette score of 0.2864 indicates limited separation, so the profiles require further validation before being used for business decisions.

Dashboard & Business Recommendations

Power BI Dashboard

The dashboard provides an interactive view of customer characteristics, purchasing behavior, subscription and discount usage, and exploratory customer segments.

1 Investigate Subscription–Discount Patterns

Validate whether the strong association reflects business rules, promotion eligibility, or a pattern in the dataset before making campaign decisions.

2 Avoid Single-Metric Customer Evaluation

Do not use recorded purchase amount, subscription status, or previous-purchase count alone as a measure of customer value.

3 Validate Exploratory Customer Profiles

Review the characteristics and potential needs of each cluster before designing differentiated marketing strategies.

4 Strengthen Customer Data Infrastructure

Collect transaction timestamps, transaction-level purchase details, promotion exposure, and customer activity history to support more reliable behavioral analysis.

Expected Business Value

A clearer view of observed customer patterns, better-informed analytical decisions, and a stronger data foundation for future customer analytics.

Dashboard & Business Recommendations

Project Limitations

No Transaction Timestamps

Actual retention, churn timing, and time-based purchasing trends cannot be measured reliably.

Limited Statistical Differences

Most tested customer groups showed no statistically significant difference in recorded purchase amount.

Exploratory Clustering

The silhouette score of 0.2864 indicates limited cluster separation.

Customer-Level Data

The dataset does not provide complete transaction histories or verified lifetime customer value.

Association Is Not Causation

Observed relationships do not establish causal effects.

Initialization Sensitivity

Cluster assignments showed sensitivity to K-Means initialization in the exploratory stability assessment.

Recommended Next Steps

Obtain transaction-level data with timestamps and stable customer identifiers.
Validate the subscription–discount relationship against business rules and source data.
Test alternative clustering configurations and validate segment usefulness.
Incorporate customer activity, product margin, and campaign response data.
Reassess findings using a more complete dataset.

Conclusion

This project demonstrates an end-to-end customer analytics workflow combining data preparation, statistical testing, exploratory segmentation, SQL analysis, and Power BI reporting. Its findings provide a starting point for further validation rather than definitive evidence of customer behavior or business impact.

Thank You

Full Project

https://github.com/nigelalessan/trendone_customer_behavior_analysis